SILTHAUL · RISK REGISTER AND HAZARD ANALYSIS

SiltHaul risk register and hazard analysis

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-03	Amish Chadha	First risk register, preliminary hazard analysis and design FMEA, written for the kit 1.9.1 upgrade

SiltHaul risk register and hazard analysis

The top risks are a rope or anchor that lets go under tension, a load that runs back if a pawl, pin or chain fails, and contamination of the crew by flood mud; the design answers them with a 3 kN shear pin limit, two ratchets, a closed guard and a sound-slab anchor rule, all checked on paper only. The highest RPN items are F1 (shear pin, 60) and F4 (tail anchors, 40), then F5, F10 and F14 at 36; each has a proof-load, checklist or site-test action for TRL 4. Method: STANDARDS section 23. Likelihood, impact, severity and occurrence are scored 1 (low) to 5 (high).

Project risk register

Table 1. Project risks.

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
P1	The output ratio of 0.57 is below 0.5 in a real trial	Assumed mud strength, floor friction and loading rates	R1 not met; option B or a longer crew needed	3	4	Design lead	Timed side-by-side trial at TRL 4; B is the fallback (SLH-OPT-001)
P2	No co-design partner is found	No agreement yet with a disaster authority or NGO	No realistic mud, room or crew to test	4	4	Project lead	Approach the first candidate named in SLH-PRB-001 (not agreed); Amish decides
P3	A local fabricator cannot reproduce the welds and fit-up	245 parts and tight cross-member fit	Frame or drum out of true; setup slips	3	3	Fabricator	Build plan with joint pictures; first build at TRL 4 (not started)
P4	The shear pin scatter is wider than 20 %	Pin steel varies by supplier	Overload limit not reliable (PR-011)	3	5	Design lead	Release test on a batch; buy pin bar from one supplier
P5	Rated shackles and a 2 t sling are not available or priced as assumed	Indicative BOM prices	Cost above the USD 2,000 target	2	3	Project lead	Quote at TRL 4; current cost USD 1,055.60
P6	Amish chooses B, A or neither and the repo must follow	Options within 10 % (SLH-OPT-001)	Re-plan of BOM, drawings and build plan for B	3	2	Project lead	Option B is modelled at D3 already; BOM and drawings follow the choice
P7	Safety claims outrun the evidence	Proof loads untested	Harm to a volunteer	2	5	Design lead	"Met on paper" only; proof-load tests before any use (SLH-BLD-001 section 6)

Preliminary hazard analysis

Table 2. Hazards.

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H1	Rope or splice parts under tension	Wear, overload, cut	Whiplash injury to a person in the line	5	2	Overload limit 3 kN; rope factor 5.4; nobody in the rope lines while the cranks turn; stop signals (BOM 22)	Medium
H2	Anchor or sling fails	Weak slab, wall used as anchor, sling over an edge	Frame or tail block flies toward the crew	5	2	Floor anchors only in a sound slab; walls never used; sling level or rising 10 degrees	Medium
H3	Load runs back when cranks are let go	Pawl missed, pin or chain fails	Spinning cranks strike a hand	4	2	Two ratchets and pawls; hands off the cranks on a stop signal	Low
H4	Crushing at the chain, sprockets or between boxes	Guard removed; fingers between boxes passing 42 mm apart	Crushed or cut finger	3	3	Closed guard; loaders keep hands and feet out from between the boxes	Medium
H5	Contaminated mud, sewage, tetanus	Cuts and wet skin in floodwater	Infection	4	4	Boots, gloves, hand washing, wound cover (Better Health Channel); the kit reduces time in the mud, not the exposure	Medium
H6	Manual handling of the kit	22.2 kg drum and bearings, 16.2 kg box	Back or shoulder strain	3	3	Heaviest part under 25 kg; two people share the drum	Low
H7	Damaged building	Weakened walls, floors, live electrics	Collapse, electric shock	5	2	Outside the design; the crew follows local advice; no wall contact (PR-007)	Medium
H8	Capstan tips or skids	Sling rising more than 10 degrees; soft ground	Frame moves toward the crew	4	2	Tipping margin 145 against 270 N m at 10 degrees; four stakes	Low
H9	Dumping a full box	Box tips forward onto feet	Foot injury	3	3	Two people on the 900 mm tipping bar; 119 N each; feet clear	Low
H10	Heat, fatigue and slips on wet mud	Hot humid work, wet footing	Collapse, falls	3	4	Rest allowance (25 % of time); staked foot boards for option B	Medium

Design FMEA (first pass, P3)

Table 3. Failure modes. RPN = severity x occurrence x detection.

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F1	Shear pin (BOM 10)	Limit rope tension to 3,000 N	Releases late (above 3 kN) or early	Rope, chain, anchors or shaft overloaded; or the crew loses the pull	Pin steel stronger than S235 or 20 % scatter; wrong pin substituted	5	3	4	60	Pins in a labelled bag, cut from S235 bar only; release test on a batch (PR-011)
F2	Ratchet wheels and pawls (BOM 8)	Hold the drum when the cranks are let go	Pawl does not engage, tooth shears	Drum runs back; cranks spin; load runs to the box	Spring lost, mud in teeth, tooth overload 4.1 kN	5	2	3	30	Two opposite ratchets; tooth bearing stress 57 MPa; release test at working pull (PR-012)
F3	Rope (BOM 16)	Carry the pull between drum and boxes	Rope parts or splice fails	Rope whips; box runs back	Abrasion on the ramp roller, cuts, UV, bend ratio 6 at the roller	5	2	3	30	Factor 5.4 at the limit; inspect before each use; replace on cut or fray
F4	Tail anchors (BOM 15)	Hold the tail block against 6 kN if a returning box jams	Wedge anchor pulls out or the slab spalls	Tail block flies toward the crew	Cracked or thin slab, drilling too close to an edge	5	2	4	40	Sound slab 100 mm or more; 0.45 of recommended loads; pull test on the anchored block (PR-007)
F5	Anchor sling (BOM 12)	Hold the capstan against the pull	Sling cut or the tree or vehicle moves	Capstan skids or tips toward the crew	Sling over a sharp edge; rising more than 10 degrees	4	3	3	36	Tree protector strap; sling level or rising 10 degrees at most; stakes stop skating
F6	Drum shaft and bearings (BOM 2, 3)	Carry the rope load into the frame	Shaft bends or a bearing seizes	Drum jams; crew loses control of the load	Mud and water in the bearing; overload at the pin limit	4	2	3	24	Shaft 144 MPa combined against 213 allowed; sealed UCP bearings with grease nipples
F7	Drum tube (BOM 2)	Carry wound rope without collapse	Tube buckles under rope pressure	Rope slackens and tangles	3 mm wall under 2.5 MPa rope pressure	4	1	4	16	Buckling factor 4.6 between rings (SLH-STM-001 S7); rings at both ends of each half
F8	Chain, sprockets (BOM 6, 7)	Transmit torque at 4:1	Chain breaks or jumps a tooth	Cranks spin free; ratchets then hold the drum	Misalignment, no lubrication, shock	3	3	3	27	Factor 4.9 at the limit (fix proposed, SLH-STM-001 S5); guard (BOM 9)

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
										keeps the chain in view and hands out
F9	Chain guard (BOM 9)	Keep fingers away from chain and sprockets	Guard removed or bent	Crushed finger at the chain	Removed for lubrication and not refitted	3	3	2	18	Two M6 screws only; briefing card; guard in the packing list
F10	Frame cross-member bolts (BOM 1, 21)	Hold the flat-pack frame together	Bolt loosens or is left out	Frame racks; drum shaft misaligns	Twelve bolts to fit on site in 3 min	4	3	3	36	Nyloc nuts; checklist in the build plan section 5; look before the first stroke
F11	Scraper box and bridles (BOM 17, 20)	Carry 86 kg of mud and its pull	Shackle pin works out, chain or lug fails	Box runs free in the room	Split pin missing, lug weld cracks	3	3	3	27	Split pin on every shackle pin; 10 mm lugs; inspect welds
F12	Tail sheave pins (BOM 13, 14)	Turn the tail rope through 180 degrees	Pin bends under a jam load	Sheave drops; rope jumps out	Jam at the limit: 138 MPa bending	3	2	3	18	25 mm S355 pins, keeper bar over both; inspect after a jam
F13	Doorway ramp (BOM 19)	Carry the boxes over the threshold	Cheek ply splits or the roller seizes	Box catches; rope snatches	Wet ply, mud in the roller bushes	2	3	2	12	Exterior plywood, nylon bushes, clean after use
F14	Option B foot boards and stakes	Give the haulers a staked foothold	Stake pulls out of soft mud	Haulers slip while pulling 756 N	Saturated soil; 189 N each on 25 mm stakes	3	3	4	36	Stakes 480 mm deep; test the pull on site before the first haul; heel cleats
F15	Option C diaphragms and valves	Pump slurry at 0.46 L/s	Valve blocked by debris; diaphragm tears	Pump stops; slurry spills	Sticks and stones in the strainer; flex fatigue at 1.7 % strain	2	4	3	24	Strainer cage; clear and test valves before each shift; supplier flex life to confirm

Actions are paper actions or TRL 4 tests; none is carried out here. Scores are the author's first pass and are to be reviewed with a co-design partner and a fabricator.